

Mahendra Kondapalli

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PROFESSIONAL SUMMARY

Generative AI Engineer with 4+ years of experience developing AI-powered applications and data pipelines that automate workflows and surface insights. Skilled in designing and deploying end-to-end solutions using Python, SQL, LangChain, AWS, Docker, and CI/CD practices to ensure reliable performance across environments. Proven track record building scalable inference pipelines, automated model evaluation frameworks, and stakeholder-driven AI tools to enhance operational efficiency.

CORE COMPETENCIES

- **Programming Languages:** Python, SQL, Javascript, C++, Scala, Bash
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, Random Forest, Hugging Face Transformers, Feature Engineering, Model Training, Hyperparameter Optimization, AI Applications Development
- **Generative AI & LLMs:** GPT-4o, Claude, Llama 3, OpenAI API, LangChain, LangGraph, RAG, Fine-Tuning, Prompt Engineering, Semantic Search, Vector Databases (Pinecone, FAISS), Agentic AI, Function Calling, LLM Evaluation
- **MLOps & Model Lifecycle:** MLflow, Kubeflow, Weights & Biases, Model Registry, Experiment Tracking, CI/CD Pipelines, Model Monitoring, Data Drift Detection, Concept Drift Detection, Model Governance, Software Development, Testing And Debugging
- **Data Engineering & Integration:** Apache Spark, Apache Airflow, Pandas, NumPy, ETL Pipelines, Data Validation, Data Quality Monitoring, Data Pipelines, Data Integration Tools, Relational Data Modeling
- **Business Intelligence & Analytics:** Data Analytics, Business Intelligence Development, Dashboards And Reports, Power BI, DAX, Semantic Models
- **Cloud & Deployment:** AWS SageMaker, EC2, ECS, EKS, S3, Lambda, Docker, Kubernetes, FastAPI, Flask, Terraform, Azure Cloud Platform, AI/ML Services
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Redis, Pinecone, FAISS
- **Monitoring & Observability:** Grafana, Evidently AI, CloudWatch, Prometheus
- **Collaboration Tools:** Git, GitHub Actions, Jira, Confluence, Agile Scrum

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Deep Learning Specialization – deeplearning.ai
- AWS Certified Cloud Practitioner

PROFESSIONAL EXPERIENCE

H2O.ai	Feb 2024 - Present
<i>Generative AI Engineer</i>	<i>San Francisco, CA</i>
• Built enterprise knowledge assistants using LangChain, vector databases, and document retrieval frameworks, enabling accurate information retrieval across large unstructured datasets. • Designed and deployed production-grade Generative AI applications leveraging LLMs, Retrieval-Augmented Generation (RAG), and semantic search architectures, improving response accuracy by 27%. • Implemented LLMops workflows including prompt evaluation, guardrails, hallucination detection, and RAG evaluation frameworks. • Developed RAG-based enterprise assistants leveraging GPT-4o, Llama 3, and Claude models through LangChain orchestration. • Developed scalable inference pipelines for transformer-based models, reducing average prediction latency by 32% and improving service reliability for enterprise users. • Implemented automated model evaluation, experiment tracking, and performance monitoring frameworks using MLflow and Weights & Biases, reducing model validation cycles by 85%. • Optimized prompt engineering strategies and retrieval pipelines through A/B testing and iterative experimentation, improving response consistency by 40%. • Deployed containerized machine learning services using Docker, Kubernetes, FastAPI, AWS SageMaker, ECS, and Azure ML Services, supporting high-volume production workloads. • Established CI/CD pipelines for machine learning model deployment, integrated automated testing and debugging, model versioning, and rollback capabilities. • Implemented model monitoring, drift detection, and observability dashboards using Evidently AI, Grafana, and CloudWatch to maintain production model quality. • Reduced hallucination rates by 18% through retrieval augmentation, prompt optimization, and fine-tuning methodologies. • Built agentic AI workflows integrating tool calling, function calling, and multi-step reasoning pipelines. • Developed robust data pipelines and data integration workflows using Apache Airflow and custom ETL scripts to prepare unstructured data for AI applications.	
Razorpay	Mar 2022 - Jul 2023

- Built feature engineering pipelines for payment transaction data, customer behavior analytics, and merchant risk assessment, processing over 2.8 million records daily.
- Developed and deployed machine learning models using XGBoost, Random Forest, and Scikit-Learn for transaction risk scoring and fraud detection, improving fraud recall by 23% while maintaining precision targets.
- Automated hyperparameter optimization using Optuna, reducing model experimentation time by 60% and improving model accuracy to 91%.
- Designed and maintained ETL pipelines using Apache Airflow, Spark, and Pandas to ingest data from multiple payment, merchant, and customer sources.
- Developed real-time prediction APIs using Flask, Docker, and AWS EC2, achieving 99.5% service availability and low-latency inference.
- Implemented A/B testing frameworks to evaluate fraud detection and risk models, enabling data-driven deployment decisions and improving business KPIs by 11%.
- Optimized relational data modeling and SQL-based feature stores in PostgreSQL, reducing query execution time by 35%.
- Applied model quantization and inference optimization techniques, reducing memory consumption by 40% and improving deployment efficiency.
- Implemented model monitoring, performance tracking, and automated retraining workflows to maintain prediction accuracy and detect model drift.
- Designed interactive dashboards and reports with Power BI, utilizing DAX calculations for detailed transaction risk visualization.

EDUCATION

University of Central Missouri	
<i>Master of Science, Computer Science</i>	USA
KL University	
<i>Bachelor of Technology, Computer Science</i>	India